

OncoImplexus Algorithm Details

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2026-01-08

OncoImplexus Algorithm Details

OncoImplexus is an integrated framework for the detection and classification of complex chromosomal rearrangements (chromoanagenesis) in cancer genomes. This document details the mathematical models and algorithmic logic for its three primary modules: **Chromothripsis**, **Chromoplexy**, and **Chromoanasyntesis**.

1. Chromothripsis Detection

The chromothripsis detection module is adapted from the statistical framework established by Cortes-Ciriano et al. (Nature Genetics, 2020) in the *ShatterSeek* algorithm. It identifies clusters of interleaved structural variants (SVs) and evaluates them against statistical criteria defining random shattering and reassembly.

1.1 SV Clustering

SVs are first clustered based on genomic proximity. Let SV_i and SV_j be two intrachromosomal SVs with breakpoints $B_i = \{b_{i1}, b_{i2}\}$ and $B_j = \{b_{j1}, b_{j2}\}$. A cluster C is defined as a connected component in a graph where nodes are SVs and an edge exists between SV_i and SV_j if their spanned intervals overlap or are contained within one another.

1.2 Statistical Criteria

For each identified cluster, the following statistical tests are performed:

A. Clustering of Breakpoints (Exponential Distribution Test) To test if breakpoints are randomly distributed (Poisson process) versus clustered, we analyze the inter-breakpoint distances. Let $X = \{x_1, x_2, \dots, x_n\}$ be the sorted genomic positions of all breakpoints in a cluster. The inter-breakpoint distances are $d_i = x_{i+1} - x_i$. Under the null hypothesis of random breakage, the distances d_i should follow an Exponential distribution:

$$f(x; \lambda) = \lambda e^{-\lambda x}$$

where λ is the rate parameter.

Standard KS Test (Legacy) In the traditional approach, λ is estimated as $\hat{\lambda} = 1/\bar{d}$, and a **Kolmogorov-Smirnov (KS) test** compares the empirical distribution against the theoretical exponential. However, this approach has a known statistical issue: when testing a composite hypothesis with estimated parameters, the standard KS test produces **conservative p-values** (systematically inflated).

Monte Carlo KS Test (Recommended, v1.1+) To address the composite null hypothesis problem, OncoImplexus implements a **Parametric Bootstrap** approach, which is the gold standard for this class of problems:

Algorithm:

1. **Compute observed statistic:** Given observed distances $\{d_1, \dots, d_n\}$, estimate $\hat{\lambda}_{obs} = 1/\bar{d}$ and compute the KS statistic:

$$D_{obs} = \sup_x |F_n(x) - F_{\hat{\lambda}_{obs}}(x)|$$

2. **Simulate null distribution:** For $b = 1, \dots, B$ (typically $B = 1000$):

- Generate n samples from $\text{Exp}(\lambda = 1)$ (scale-invariant due to KS property)
- Estimate $\hat{\lambda}^{(b)} = 1/\bar{d}^{(b)}$ from simulated data
- Compute $D^{(b)} = \sup_x |F_n^{(b)}(x) - F_{\hat{\lambda}^{(b)}}(x)|$

3. **Compute empirical p-value:**

$$p = \frac{\#\{D^{(b)} \geq D_{obs}\} + 1}{B + 1}$$

The “+1” correction ensures $p > 0$ and provides a conservative estimate.

Mathematical Justification:

The KS statistic for exponential distributions is scale-invariant:

$$D(\{d_i\}, \lambda) = D(\{d_i/c\}, c\lambda)$$

This allows simulation with $\lambda = 1$ regardless of the true rate, making the bootstrap computationally efficient.

Computational Complexity:

Breakpoints (n)	Simulations (B)	Typical Runtime
10	1000	< 0.1 sec
100	1000	< 0.5 sec
500	1000	~ 1-2 sec

Note: OncoImplexus (v1.0.0+) implements a critical fix in this test by correctly extracting and ordering breakpoint coordinates, resolving a coordinate-swap bug found in earlier frameworks that led to significant false-negative rates in complex cancer genomes.

B. Randomness of Fragment Joins (Bernoulli/Multinomial Test) Chromothripsis involves the random rejoining of DNA fragments. This implies that the four types of fragment joins (resulting in different SV types) should occur with equal probability. The four join types (SV types) are: 1. **Deletion-like (DEL):** (+, -) orientation 2. **Duplication-like (DUP):** (-, +) orientation 3. **Head-to-Head Inversion (h2hINV):** (+, +) orientation 4. **Tail-to-Tail Inversion (t2tINV):** (-, -) orientation

We use a **Chi-squared Goodness-of-Fit test** (or exact Multinomial test) to test the null hypothesis:

$$P(\text{DEL}) = P(\text{DUP}) = P(\text{h2hINV}) = P(\text{t2tINV}) = 0.25$$

Significant deviation (low p-value) suggests non-random mechanisms (e.g., selection pressure), though “High Confidence” chromothripsis typically requires adhering to this random profile.

C. Chromosomal Enrichment We test if the chromosome containing the cluster has significantly more SVs than expected by chance, given its length relative to the genome. Let N_{chr} be the number of SVs on the candidate chromosome, and N_{total} be the total SVs in the sample. Let L_{chr} and L_{genome} be the effective lengths. The expected probability is $p = L_{chr}/L_{genome}$. A **Binomial Test** calculates the probability of observing $\geq N_{chr}$ events:

$$P(X \geq N_{chr}) = \sum_{k=N_{chr}}^{N_{total}} \binom{N_{total}}{k} p^k (1-p)^{N_{total}-k}$$

D. Copy Number Oscillations Chromothripsis typically produces oscillating copy number (CN) states (e.g., between 1 and 2 copies). We calculate the maximum number of consecutive oscillating CN segments. Let $C = (c_1, c_2, \dots, c_m)$ be the sequence of total copy numbers for adjacent segments in the region. An oscillation is defined as a transition $c_i \rightarrow c_{i+1}$ where $|c_i - c_{i+1}| \leq 1$. The metric M_{osc} counts the length of the longest sub-sequence satisfying oscillating conditions (2-state or 3-state).

2. Chromoplexy Detection (v3.0)

OncoImplexus introduces a novel **Graph-Based Chaining Algorithm** (v3.0) to detect chromoplexy, overcoming limitations in previous methods that failed to connect chains when local rearrangements or deletions occurred at breakpoints.

2.1 Graph Construction

We construct a graph $G = (V, E)$ where nodes V represent genomic breakpoints. The edge set E is the union of three distinct edge types: $E = E_{tra} \cup E_{adj} \cup E_{del}$.

A. Translocation Edges (E_{tra}) Represent observed inter-chromosomal structural variants. For a translocation between $chrA : pos1$ and $chrB : pos2$:

$$(v_{A,1}, v_{B,2}) \in E_{tra}$$

Weight: $w_{tra} = 1.0$

B. Genomic Adjacency Edges (E_{adj}) Represent the physical proximity of breakpoints on the *same* chromosome, allowing the algorithm to “walk” along the chromosome to find the next translocation. For two breakpoints v_i, v_j on the same chromosome with distance $d_{ij} = |pos_i - pos_j|$:

$$(v_i, v_j) \in E_{adj} \iff d_{ij} < D_{max_adj}$$

where D_{max_adj} is the adjacency threshold (default 10Mb). The edge weight decays exponentially with distance:

$$w_{adj}(d_{ij}) = e^{-d_{ij}/\sigma}$$

where σ is a decay scale (default 5Mb).

C. Deletion Bridges (E_{del}) Represent inferred deletions that connect two breakpoints (e.g., a “bridge” formed by the loss of the segment between them). A deletion bridge is identified if a CNV deletion segment S_{del} spans or is adjacent to breakpoints v_i, v_j .

$$(v_i, v_j) \in E_{del}$$

Confidence score C_{bridge} is calculated based on distance, deletion size, and copy number:

$$C_{bridge} = 0.5 \cdot e^{-d/200kb} + 0.3 \cdot S_{size} + 0.2 \cdot e^{-(CN)/1}$$

2.2 Chain Discovery (Backtracking)

We search for chains (paths) in G that satisfy chromoplexy criteria using a backtracking algorithm (Depth-First Search with constraints). A path $P = (v_1, v_2, \dots, v_k)$ is a valid chromoplexy chain candidate if: 1. **Length:** $N_{tra}(P) \geq N_{min_tra}$ (Minimum translocations, default 3). 2. **Complexity:** $N_{chr}(P) \geq N_{min_chr}$ (Minimum chromosomes involved, default 3). 3. **Graph Structure:** The path traverses a sequence of translocation edges interleaved with adjacency/deletion edges.

The search space is pruned using: * **max_depth:** Maximum chain length. * **max_neighbors:** Limit on adjacency edges per node (optimization for complex graphs).

2.3 Evaluation and Scoring

Each candidate chain is scored:

$$Score_{combined} = 0.3 \cdot S_{CN} + 0.3 \cdot S_{complexity} + 0.2 \cdot F_{del} + 0.2 \cdot S_{stat}$$

Where: * $S_{CN} = e^{-\text{mean}(|CN-2|)/2}$: Copy number stability score. * $S_{complexity}$: Normalized score based on number of chromosomes and translocations. * F_{del} : Fraction of edges that are deletion bridges.

3. Chromoanasythesis Detection

This module detects replication-based mechanisms (Fork Stalling and Template Switching - FoSTeS / MMBIR), characterized by local clustering of SVs, copy number gradients, and linked breakpoint chains.

3.1 Copy Number Gradient Detection

Chromoanasythesis often produces a stepwise increase in copy number. We detect this using a sliding window approach. For a window of CN segments W , we calculate: 1. **Spearman Correlation (ρ)**: Correlation between segment index and Total CN.

$$\rho = \text{cor}(\text{rank}(\text{index}), \text{rank}(\text{CN}))$$

2. **Total Variation (V)**: Measures local instability (“jumpiness”).

$$V = \frac{\sum |CN_{i+1} - CN_i|}{\text{mean}(\text{CN})}$$

A region is flagged if $\rho > \rho_{thresh}$ (default 0.4) or if V is sufficiently high with a positive trend.

3.2 SV Topology Analysis

We analyze the topology of SVs within the flagged regions to distinguish FoSTeS from Breakage-Fusion-Bridge (BFB) cycles.

A. Linked SV Chains (FoSTeS Signature) FoSTeS involves serial template switching. We detect this by finding chains of “linked” SVs. Two SVs, SV_i and SV_j , are **linked** if any of their breakpoints are within a small distance δ (default 2kb, representing the replication fork jumping distance).

$$\text{Linked}(SV_i, SV_j) \iff \min_{b \in B_i, b' \in B_j} |b - b'| < \delta$$

We construct an adjacency graph of SVs and find connected components (chains).

$$L_{chain} = \max(\text{size}(\text{ConnectedComponents}))$$

High L_{chain} (≥ 3 or 4) is a strong indicator of chromoanasythesis.

B. Fold-back Inversions (BFB Signature) BFB cycles are characterized by “fold-back” inversions (very short span inversions).

$$SV \in \text{FoldBack} \iff SV_{type} \in \{h2hINV, t2tINV\} \wedge |pos2 - pos1| < 20kb$$

High density of fold-backs penalizes the chromoanasythesis score (suggesting BFB instead).

3.3 Classification

Final classification relies on a weighted complexity score:

$$Score = 0.2 \cdot S_{grad} + 0.2 \cdot S_{var} + 0.2 \cdot S_{dup} + 0.4 \cdot S_{chain}$$

Events are classified as “Likely” or “Possible” based on this score and the absence of BFB features.

4. DNA Repair Mechanism Inference

OncoImplexus analyzes the base-level sequence features at structural variant junctions to infer the operative DNA double-strand break (DSB) repair pathways.

4.1 Sequence Extraction and Orientation

To detect microhomology accurately, reference sequences flanking each breakpoint are extracted and re-oriented based on the SV strand.

Let S_1 and S_2 be the reference sequences at the two breakpoints. They are transformed into: * **Kept Sequence** (s_{kept}): The sequence retained in the rearranged genome, oriented facing the junction. * **Discarded Sequence** (s_{disc}): The sequence “lost” or “overlapped” during the rearrangement, starting exactly from the breakpoint position.

This normalization ensures that regardless of the original SV type (DEL, DUP, INV), the junction interface always follows a unified coordinate system where the junction-adjacent base of Side 1 is at the start of its discarded fragment, and Side 2 is at the end.

4.2 Microhomology Detection (Unified Suffix-Prefix Algorithm)

Microhomology is defined as an identical sequence present at both breakpoint interfaces that is represented only once in the resulting junction.

We identify the maximum length k such that:

$$\text{prefix}(s_{1,disc}, k) == \text{suffix}(s_{2,disc}, k)$$

where $k \leq \max_microhomology$ (default 25 bp).

The match must be **continuous** starting from the junction. Any mismatch terminates the search, ensuring that only biologically relevant microhomology is reported.

Methodological Note: Continuous Microhomology Assumption

OncoImplexus implements a **strict continuous match** requirement: microhomology must extend without gaps from the exact junction breakpoint. This design choice is based on:

1. **Biological relevance:** True MMEJ-mediated repair utilizes contiguous homology for annealing
2. **Reduced false positives:** Allowing gaps would detect spurious matches in repetitive sequences
3. **Consistency with major tools:** This approach aligns with BreakDancer, GRIDSS, and other established SV callers

Alternative definitions in literature: - Some studies report “gapped microhomology” allowing 1-2 bp mismatches - Extended microhomology with internal insertions (templated insertions)

Users requiring alternative definitions should post-process the extracted sequences (\$sequences output) with custom matching algorithms.

4.3 Classification Criteria

The inferred repair mechanism is determined by the length of microhomology (L_{MH}) and the type of SV:

Mechanism	Criteria	Biological Significance
NHEJ	$L_{MH} \in \{0, 1\}$ bp	Classic Non-Homologous End Joining; often associated with Chromothripsis.
MMEJ	$2 \leq L_{MH} \leq 20$ bp	Microhomology-Mediated End Joining (or Alt-EJ); suggests error-prone repair.
SSA	$L_{MH} > 20$ bp	Single-Strand Annealing; involves long homologous repeats.
MMBIR	$2 \leq L_{MH} \leq 5$ bp AND SV = DUP	Microhomology-Mediated Break-Induced Replication; a hallmark of Chromoanasythesis.

4.4 Insertion Detection

If no microhomology is found, the algorithm checks for non-templated insertions (L_{ins}) using the VCF INSSEQ or ALT fields. The presence of small insertions (typically <10 bp) further supports an NHEJ repair origin.

5. Fusion Gene Prediction

This module identifies potential oncogenic fusion genes and disruptions created by complex chromosomal rearrangements.

5.1 Inter-genic Structural Variant Scanning

Let $G = \{g_1, g_2, \dots, g_m\}$ be a set of gene models with genomic intervals $I_g = [start_g, end_g]$ and strands $S_g \in \{+, -\}$. An SV with breakpoints $\{b_1, b_2\}$ is identified as a potential fusion source if:

$$\exists g_i, g_j \in G : b_1 \in I_{g_i} \wedge b_2 \in I_{g_j} \wedge g_i \neq g_j$$

5.2 Orientation Compatibility Algorithm

To support a functional 5' to 3' chimeric transcript, the genomic join must preserve the transcription polarity. Let $s_1, s_2 \in \{+, -\}$ be the strands of the SV breakpoints (representing the side retained).

A gene g acts as a valid **5' Partner (Head)** if: * $S_g = +$ and $s = +$ (Junction is to the right of the breakpoint) * $S_g = -$ and $s = -$ (Junction is to the left of the breakpoint)

A gene g acts as a valid **3' Partner (Tail)** if: * $S_g = +$ and $s = -$ (Junction is to the left of the breakpoint) * $S_g = -$ and $s = +$ (Junction is to the right of the breakpoint)

5.3 Classification

- **Canonical Fusion:** Both 5' and 3' partner conditions are met by the two involved genes. This orientation allows the mRNA polymerase to read across the junction into the 3' partner.
- **Non-canonical / Disruption:** The orientation is incompatible with a functional 5'→3' transcript (e.g., Head-to-Head join). Such events are flagged as gene disruptions rather than productive fusions.

6. Statistical Considerations and Limitations

This section provides critical information about the statistical assumptions underlying OncoImplexus and their practical implications.

6.1 Exponential Distribution Test Caveats

Composite Null Hypothesis Problem The Kolmogorov-Smirnov (KS) test used in chromothripsis detection tests the null hypothesis that inter-breakpoint distances follow an exponential distribution. However, the rate parameter λ is **estimated from the sample** rather than specified a priori.

When testing a composite hypothesis (distribution family with unknown parameters), the standard KS test is **conservative**, meaning: - The actual Type I error rate is lower than the nominal α - The test has reduced power to reject false null hypotheses - P-values are systematically inflated

Mathematical Basis:

Let $F_n(x)$ be the empirical CDF and $F_{\hat{\lambda}}(x)$ be the exponential CDF with estimated rate. The KS statistic:

$$D_n = \sup_x |F_n(x) - F_{\hat{\lambda}}(x)|$$

Under the composite null, D_n follows a modified distribution (Lilliefors distribution), not the standard Kolmogorov distribution. Critical values from standard tables are **too large**.

Practical Implication:

A “non-significant” p-value (e.g., $p > 0.05$) does not necessarily mean distances are exponentially distributed. Consider this as supportive rather than definitive evidence.

OncoImplexus Solution (v1.1+):

OncoImplexus addresses this problem using **Parametric Bootstrap** (Monte Carlo simulation), which is the gold standard for goodness-of-fit testing with estimated parameters. See Section 1.2A for the detailed algorithm.

This approach: - Produces correctly calibrated p-values - Requires no external dependencies - Is computationally efficient for typical breakpoint counts

Legacy Alternatives (for reference): 1. **Anderson-Darling test** - more sensitive to tail deviations (requires `gofest` package) 2. **Lilliefors test** - corrected critical values (limited availability for exponential)

6.2 Chi-Squared Test Assumptions

The fragment joins randomness test assumes: 1. **Independence**: Each SV join is independent 2. **Expected frequency ≥ 5** : All four categories (DEL, DUP, h2hINV, t2tINV) should have expected count ≥ 5

When expected frequencies are small:

$$n \cdot p_i = n/4 < 5 \implies n < 20$$

For clusters with fewer than 20 SVs, the chi-squared approximation may be unreliable.

Recommendation:

For small clusters ($n < 20$), interpret fragment joins p-values with caution or use Fisher’s exact test.

6.3 Copy Number Oscillation Definition

The oscillation metric uses a strict definition:

$$|c_i - c_{i+1}| \leq 1$$

This assumes **integer copy number values**. In practice, CNV callers output continuous estimates with measurement noise. The implementation rounds or truncates values, which may affect results.

Considerations: - Floating-point CN values (e.g., 1.8, 2.1) are treated as distinct from 2.0 - Minor noise can artificially break oscillation chains - Consider pre-rounding CN values if this is a concern

6.4 Correlation Threshold for Chromoanasythesis

The gradient detection uses Spearman rank correlation with a fixed threshold ($\rho_{thresh} = 0.4$). This threshold was chosen based on empirical observations but: - Not validated across all cancer types - Does not account for the number of segments (correlation significance depends on sample size) - May require adjustment for different CNV callers

Segment Count Effect on Correlation Significance:

Segments (n)	Critical ρ ($\alpha=0.05$)	Critical ρ ($\alpha=0.01$)
5	0.90	1.00
10	0.65	0.79
20	0.45	0.57
50	0.28	0.36

For small segment counts, even high correlations may not be statistically significant.

7. Integrated Detection Workflow

OncoImplexus provides an integrated `detect_chromoanagenesis()` function that orchestrates all three detection modules and resolves overlapping events.

7.1 Execution Pipeline

Input: `SV.sample`, `CNV.sample`, `genome`

```

|
|-> [1] Chromothripsis Detection
|     |-> Intra-chromosomal SV clustering
|     |-> Statistical criteria evaluation
|     |-> Classification (High/Low Confidence)
|
|-> [2] Chromoplexy Detection
|     |-> Inter-chromosomal SV extraction
|     |-> Graph construction (TRA + ADJ + DEL edges)
|     |-> Chain discovery (backtracking)
|     |-> Chain evaluation and scoring
|
|-> [3] Chromoanasythesis Detection
|     |-> CN gradient analysis
|     |-> SV topology analysis (chains, fold-backs)
|     |-> BFB vs FoSTeS discrimination
|
v
[4] Mixed Mechanism Classification
|-> Spatial overlap detection
|-> Chromosome-level assignment
|-> Sample-level integration

```

| -> Global FDR correction across mechanisms

7.2 Multiple Testing Correction

OncoImplexus performs multiple statistical tests across chromosomes and mechanisms. To control the false discovery rate (FDR), we apply the Benjamini-Hochberg procedure:

Let $p_1, p_2, \dots, p_m$ be the sorted p-values from all tests. The adjusted p-value (q-value) is:

$$q_i = \min \left(p_i \cdot \frac{m}{i}, q_{i+1} \right)$$

Scope of FDR Correction: - Chromothripsis: Per-chromosome tests (fragment joins, exponential, enrichment) - Chromoplexy: Per-chain significance tests - Cross-mechanism: Global FDR combining chromothripsis and chromoplexy p-values

7.3 Conflict Resolution

When multiple mechanisms are detected on the same chromosome or region:

1. **Non-overlapping events:** Both mechanisms are reported
2. **Overlapping events:** Classified as “Mixed” with dominant mechanism annotation
3. **Dominance determination:** Based on:
 - Confidence scores
 - Number of supporting SVs
 - Statistical significance

8. Algorithm Limitations

8.1 Input Data Requirements

Requirement	Chromothripsis	Chromoplexy	Chromoanasythesis
Intra-chr SVs	Required	Not used	Required
Inter-chr SVs	Optional	Required	Optional
CNV segments	Required	Recommended	Required
Breakpoint strands	Required	Required	Required
BSgenome (repair)	Optional	Optional	Optional

8.2 Known Limitations

1. **SV Caller Dependency:** Results sensitive to SV calling accuracy
 - Missing calls -> false negatives
 - False calls -> potential false positives
 - Imprecise breakpoints -> reduced repair mechanism accuracy
2. **CNV Segmentation Effects:**
 - Over-segmentation -> inflated oscillation counts
 - Under-segmentation -> missed events
 - Resolution affects gradient detection
3. **Computational Complexity:**
 - Chromoplexy chain search is $O(b \cdot d^n)$ where b = branches, d = depth, n = nodes
 - Highly connected graphs (>500 inter-chr SVs) may require parameter tuning
4. **Reference Genome Requirement:**
 - Repair mechanism inference requires BSgenome object
 - Chromosome naming must match (chr1 vs 1)

8.3 Edge Cases

Scenario	Behavior	Recommendation
No SVs detected	Returns empty results	Check SV caller output
Only inter-chr SVs	Chromothripsis/CS skipped	Expected behavior
Very large SV counts (>1000)	Potential timeout	Reduce <code>max_path_search</code>
Mismatched chr names	Analysis fails	Standardize naming
Empty CNV data	Partial analysis only	Provide CNV data for complete analysis

9. Performance Characteristics

9.1 Time Complexity

Module	Complexity	Typical Runtime (100 SVs)
Chromothripsis	$O(n^2)$ for clustering	< 5 seconds
Chromoplexy	$O(k \cdot d^m)$ for chain search	5-30 seconds
Chromoanasythesis	$O(s \cdot w)$ for sliding window	< 5 seconds

Where: n = SVs, k = start nodes, d = branching factor, m = max depth, s = segments, w = window sizes.

9.2 Memory Usage

Primary memory consumers: - Adjacency matrix for SV clustering: $O(n^2)$ - Chromoplexy graph edges: $O(n \cdot k)$ where k = max neighbors - Path storage during backtracking: $O(depth \cdot paths)$

Recommendation: For samples with >500 SVs, consider: - Reducing `max_path_search` - Limiting `adjacency_distance` - Using `max_neighbors` parameter

References

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